

Maria Andrade

Hydrology, Tucson, USA

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EDUCATION

Ph.D. in Hydrology, minor in Remote Sensing

JANUARY 2025–PRESENT

University of Arizona

Tucson, United States of America

Financial support: U.S. Department of Energy (DOE). Supporting research on hydrological processes and groundwater–surface water interactions to assess remediation effectiveness and protect critical habitats in the Colorado River under current and future climate conditions. Supervisor: Peter A. Troch.

M.Sc. in Hydraulics and Sanitation

FEBRUARY 2023–DECEMBER 2024

University of Sao Paulo

Sao Carlos, Brazil

Thesis: “Deterministic and probabilistic methods for assessing the impact of dam breaks via hydrodynamic modeling.” DOI: [10.11606/D.18.2025.tde-21032025-085649](https://doi.org/10.11606/D.18.2025.tde-21032025-085649). Honorable Mention, Ministry of Defense/CAPES National Defense Theses and Dissertations Competition. Supervisor: Eduardo Mario Mendonço.

B.Sc. in Civil Engineering

2016–2021

Federal University of Pernambuco

Recife, Brazil

Senior thesis: “Assessing the socioeconomic benefits of the Agreste Adduction System in Pernambuco, Brazil.” Supervisor: Suzana Maria Gico Lima Montenegro.

RESEARCH INTERESTS

- Hydrology; hydraulics; optimization and analysis of large-scale sensing systems; machine learning.
- Remote sensing; surface water modeling; groundwater modeling; dam-break modeling; climate change modeling.

RESEARCH EXPERIENCE

Graduate Research Assistant

JANUARY 2025–PRESENT

University of Arizona, Department of Hydrology and Atmospheric Sciences

Tucson, United States of America

Research on hydrological processes and groundwater–surface water interactions to assess remediation effectiveness and protect critical habitats in the Colorado River under current and future climate conditions.

Graduate Research Assistant

FEBRUARY 2023–DECEMBER 2024

University of Sao Paulo, Department of Hydraulics and Sanitation

Sao Carlos, Brazil

Research on probabilistic dam-break modeling, including Monte Carlo flood simulations and uncertainty assessment using 2D hydrodynamic models. Supervisor: Eduardo Mario Mendonço.

Undergraduate Research Assistant

AUG. 2018–JUL. 2020

Federal University of Pernambuco

Recife, Brazil

Research assistant on managed aquifer recharge and water-resources projects. Supervisor: Suzana Maria Gico Lima Montenegro.

RESEARCH PROJECTS

BRAMAR

Strategies and technologies for mitigating water scarcity in Northeast Brazil. International Brazil–Germany cooperation focused on advanced effluent treatment, reuse, managed aquifer recharge, and integrated water-resources management in case studies in Northeast Brazil. Project publication: [10.3390/w12041072](https://doi.org/10.3390/w12041072).

DIGIRES

Digital-enabled green infrastructure for sustainable water resources management. Project focused on managed aquifer recharge as a green, nature-inspired water-infrastructure strategy, combining ICT tools and citizen observations for the design and implementation of artificial recharge. Website: digires.inowas.com/about.

SMART Control **SMART Framework for Real-Time Monitoring and Control of Subsurface Processes in Managed Aquifer Recharge Applications.** Development of an online real-time monitoring and control system for aquifer recharge, including a web platform for control, modeling, forecasting, risk assessment, and management. Website: smart-control.inowas.com/the-project.

PEER-REVIEWED JOURNAL PUBLICATIONS AND MANUSCRIPTS

2026

6. Gomes Jr., M. N., and **Castro, M. A. R. A.** *Continental-scale XGBoost Framework for Multi-Hazard Susceptibility and Socioeconomic Exposure in South America.* **Sustainable Cities and Society**. DOI: [10.1016/j.scs.2026.107858](https://doi.org/10.1016/j.scs.2026.107858).
5. Sánchez, M. H., Gomes Jr., M. N., Castillo, L. M., **Castro, M. A.**, and Mendiondo, E. M. (2026). *Assessment of urban vulnerability to floods under current and future demographic and climate scenarios.* **International Journal of Disaster Risk Reduction**, 106337. DOI: [10.1016/j.ijdr.2026.106337](https://doi.org/10.1016/j.ijdr.2026.106337).
4. Brandão, A. R., **Castro, M. A.**, Sánchez, M. H., Gomes Jr., M. N., Uchôa, J. G. S., Vaz, I. C., et al. (2026). *Flood Event Escalation and Urban Drainage Design Implications Under Nonstationary Rainfall in São Paulo State, Brazil.* **Water**, 18(5), 561. DOI: [10.3390/w18050561](https://doi.org/10.3390/w18050561).
3. **Castro, M. A. R. A.**, Gomes, M. N., and Montenegro, S. M. G. L. (2026). *Assessing the socioeconomic benefits of the Agreste Adduction System in Pernambuco, Brazil.* **RBRH**, 31, e04. DOI: [10.1590/2318-0331.312620250154](https://doi.org/10.1590/2318-0331.312620250154).
2. Gomes Jr., M. N., **Castro, M. A. R. A.**, Castillo, L. M. R., Sánchez, M. H., Giacomoni, M. H., de Paiva, R. C. D., and Bates, P. D. (2026). *Spatio-temporal performance of 2D local inertial hydrodynamic models for urban drainage and dam-break applications.* **Journal of Hydrology**, 665, 134661. DOI: [10.1016/j.jhydrol.2025.134661](https://doi.org/10.1016/j.jhydrol.2025.134661).

2025

1. Gomes, M. N., Jalihal, V., **Castro, M.**, and Mendiondo, E. M. (2025). *Exploring the impact of rainfall temporal distribution and critical durations on flood hazard modeling.* **Natural Hazards**, 121(9), 10989–11012. DOI: [10.1007/s11069-025-07186-3](https://doi.org/10.1007/s11069-025-07186-3).

2024

1. Gomes Jr., M. N., **Castro, M. A. R. A.**, da Silva, P. G. C., Giacomoni, M. H., and Mendiondo, E. M. (2024). *Increasing flood awareness through dam-break serious games.* **International Journal of Disaster Risk Reduction**. DOI: [10.1016/j.ijdr.2024.104543](https://doi.org/10.1016/j.ijdr.2024.104543).

PUBLICATIONS UNDER REVIEW OR UNDER PREPARATION

8. Gomes Jr., M. N., **Castro, M.**, Ballarin, A., Troch, P., Almagro, A., and Oliveira, P. T. S. *Levels and Indicators for Monitoring Inundation Across Rivers (LIMIAR): Hydraulic Flood Reconstruction in Brazil.* **Hydrology and Earth System Sciences** (under preparation).
7. Gomes Jr., M. N., **Castro, M. A. R. A.**, Nguyen, A., Wang, A., and Gorelick, S. M. *Global Inequalities in Urban Hazard Risk.* **Nature Communications** (under preparation).
6. **Castro, M. A. R. A.**, Gomes Jr., M. N., Mendiondo, E. M., and Troch, P. A. *Municipal hazard diversity and compound disaster impacts across Brazil.* **npj Natural Hazards** (under preparation).
5. **Castro, M. A. R. A.**, Hazenberg, P., Duan, J., and Troch, P. A. *Hydraulic and thermal controls on habitat suitability, persistence, and connectivity in river corridors.* **Ecohydrology** (under preparation).
4. Uchôa, J. G. S. M., Brandão, A. R. A., Ballarin, A. S., Leite, C. C. O., Pantarotto, H. S., Sánchez, M. H., **Castro, M. A. R. A.**, Dutra Júnior, S. A., Viana, V. N. R., Scanlon, B. R., Wendland, E. C., Zipper, S., and Oliveira, P. T. S. *Conjunctive water use management as a pathway to water resilience in Brazil.* Submitted to **Nature Communications**.
3. **Castro, M. A. R. A.**, Gomes Jr., M. N., Rapallo, L. M., and Mendiondo, E. M. *Assessing the trade-offs between full momentum and local-inertial hydrodynamic models under dam-break scenarios.* Submitted to **Environmental Modeling & Assessment**. DOI: [10.21203/rs.3.rs-7160953/v1](https://doi.org/10.21203/rs.3.rs-7160953/v1).
2. **Castro, M. A. R. A.**, Gomes Jr., M. N., and Mendiondo, E. M. *Probabilistic Dam Break Flood Mapping via Monte Carlo Simulations using a 2D Local-Inertial Model.* Submitted to **Journal of Hydrology**. DOI: [10.22541/es-soar.175242171.16927854/v1](https://doi.org/10.22541/es-soar.175242171.16927854/v1).

1. Sánchez, M. H., Gomes Jr., M. N., Castillo, L. M. R., **Castro, M. A. R. A.**, and Mendiondo, E. M. *An adaptive fuzzy logic control approach for flood risk management in a poorly gauged urban watershed: A proof of concept.* (under preparation).

CONFERENCE PAPERS AND PRESENTATIONS

12. **Castro, M.**, Hazenberg, P., and Troch, P. Dynamic Hydrothermal Controls on Nursery Habitat Persistence and Connectivity in a Regulated Colorado River Corridor. Submitted to **AGU Annual Meeting 2026**.
11. **Castro, M.**, Hazenberg, P., Parras, D., and Troch, P. Quantifying shallow groundwater–river recharge and flow variability to support habitat assessment near the Moab UMTRA site. **WM2026: 52nd Annual Waste Management Symposium**, Phoenix, AZ, United States, Mar. 2026. Poster No. 26670.
10. **Castro, M.**, Rapalo, L., Silva, P., and Mendiondo, E. (2024). Analysis of the potential direct and indirect impacts of dam failures on storage services for multiple purposes (No. EGU24-13615). **Copernicus Meetings**, Apr. 2024.
9. **Castro, M. A. R. A.**; Gomes Jr., M. N.; Sanchez, M. H.; Filho, P. B. S.; Mendiondo, E. M. Risk Index Review Applied to Dam Break Simulation in Brazil. **XXV Simpósio Brasileiro de Recursos Hídricos**, Aracaju, Brazil, Nov. 2023. In Portuguese.
8. Sanchez, M. H.; Gomes Jr., M. N.; **Castro, M. A. R. A.**; Mendiondo, E. M. River flow forecasting methods: A review. **XXV Simpósio Brasileiro de Recursos Hídricos**, Aracaju, Brazil, Nov. 2023.
7. Brandão, A. R. A.; Barros, A. A.; Poloni, J. S.; Areias, J. L. A.; Cantarelli, J. C. P.; **Castro, M. A. R. A.**; Benso, M. R.; Sanchez, M. H.; Filho, P. B. S.; Mendiondo, E. M. Economic Impact Assessment for Enhancing Resilience of Drainage Structures under Changing Climate Conditions in Brazilian Cities. **XXV Simpósio Brasileiro de Recursos Hídricos**, Aracaju, Brazil, Nov. 2023.
6. **Castro, M. A. R. A.**; Barros, J. C. F.; Silva, E. P.; Casiuch, M. Effects of the Implementation of the Agreste Pipeline System on Water Supply in the city of Santa Cruz do Capibaribe–PE. **IV Simpósio da Bacia Hidrográfica do Rio São Francisco**, Belo Horizonte, Brazil, Sep. 2022. In Portuguese.
5. Barros, J. C. F.; **Castro, M. A. R. A.**; Silva, E. P.; Casiuch, M. Impact of the Implementation of the Agreste Pipeline System on Water Supply in the city of Bezerros–PE. **IV Simpósio da Bacia Hidrográfica do Rio São Francisco**, Belo Horizonte, Brazil, Sep. 2022. In Portuguese.
4. Silva, E. P.; Barros, J. C. F.; **Castro, M. A. R. A.**; Casiuch, M. Impact of the Implementation of the Agreste Pipeline System on Water Supply in the city of Arcoverde–PE. **IV Simpósio da Bacia Hidrográfica do Rio São Francisco**, Belo Horizonte, Brazil, Sep. 2022. In Portuguese.
3. Fernandes, L. A.; Albuquerque, C. B.; **Castro, M. A. R. A.**; Silva, A. N.; Paiva, A. L. R.; Montenegro, S. M. G. L. Monitoring the Piezometric Level of the Cabo Aquifer in the Coastal Region of Recife–PE. **XVI SRHNe / 15th SILUSBA**, Caruaru, Brazil, Nov. 2022. In Portuguese.
2. Montenegro, A. A. A.; Carvalho, A. A.; Montenegro, H.; Costa, R. S.; Silva, J. L. D. F.; **Castro, M. A. R. A.** Application of Low-Cost Automated Systems in Teaching and Research in Water Resources and Agricultural Engineering. **XVI SRHNe / 15th SILUSBA**, Caruaru, Brazil, Nov. 2022. In Portuguese.
1. Fernandes, L. A.; Conrad, A.; Montenegro, S. M. G. L.; Paiva, A. L. R.; **Castro, M. A. R. A.** Application and evaluation of an advanced aquifer storage and recovery pilot system in Recife, Brazil. **10th International Symposium on Managed Aquifer Recharge**, Madrid, Spain, May 2019.

LEADERSHIP AND UNIVERSITY SERVICE

Social Chair

Hydrology and Atmospheric Sciences Student Association, University of Arizona

Organized departmental events and collaborated with students, faculty, and staff to strengthen engagement and community within Hydrology and Atmospheric Sciences.

AUG. 2025–MAY 2026

TEACHING AND PROFESSIONAL EXPERIENCE

Water Resources Engineer

AUG. 2020–MAY 2024

TPF Engineering

Worked on water resources and sanitation projects within the São Francisco River Integration Project, the largest water infrastructure project in Brazil. The role focused on hydraulic modeling, design development, and transient analysis of pressurized pipeline systems supplying rural communities in semi-arid Northeast Brazil for human consumption and irrigated agriculture. Main responsibilities included:

- Developing steady-state and extended-period hydraulic models for transmission mains, pumping stations, reservoirs, and pressurized distribution networks.
- Performing transient (water hammer) analyses for pump start-up and shutdown, power failures, and valve operations, and defining surge protection devices such as air valves, pressure relief valves, and surge tanks.
- Sizing pipelines and electromechanical equipment based on hydraulic performance, energy efficiency, and long-term operational sustainability.
- Contributing to commissioning procedures and validation of system performance, including flow, pressure, and distribution efficiency.

Graduate Teaching Assistant, Field Hydrology

JAN.–MAY 2026

University of Arizona

Undergraduate Teaching Assistant, Water Resources Engineering

FEB.–JUL. 2021

Federal University of Pernambuco

Undergraduate Teaching Assistant, Hydrology

AUG.–DEC. 2020

Federal University of Pernambuco

Teaching Assistant, Mathematics and Chemistry

MAR.–JUL. 2020

Enactus

Volunteering

For high school students.

Teacher

MAR.–NOV. 2015

Mangueira da Torre Project

Volunteering

Teaching classes for children and students from favelas in Northeast Brazil.

AWARDS, FELLOWSHIPS, AND SCHOLARSHIPS

2026	Thomas R. Brown Foundation Distinguished Graduate Fellowship in Integrative Science. Awarded for doctoral study in Hydrology and Atmospheric Sciences based on academic excellence, intellectual merit, and potential to advance knowledge. Provides a \$7,500 annual fellowship.
2025	Honorable Mention, MD/CAPES National Defense Theses and Dissertations Competition (Master's Category). Awarded for the thesis "Deterministic and probabilistic methods for assessing the impact of dam breaks via hydrodynamic modeling," University of São Paulo, São Carlos. Links: EESC/USP news ; CAPES result .
2025–present	Ph.D. degree scholarship, University of Arizona, Tucson, United States of America. Funding support: approximately \$105,600 USD.
2023–2024	M.Sc. degree scholarship, University of Sao Paulo, Sao Carlos, Brazil. Funding source: CAPES. Estimated support: approximately R\$ 46,000. Funding source .
Aug. 2018–Jul. 2020	Undergraduate Research Assistant scholarship, Science and Technology Support Foundation (FACEPE), Recife, Pernambuco. Funding support: approximately R\$ 9,600. Funding table .
Aug. 2016–May 2022	B.Sc. degree free tuition by state university scholarship, Federal University of Pernambuco, Recife, Pernambuco. Estimated tuition support: approximately R\$ 105,000.00.

SOFTWARE AND PROGRAMMING SKILLS

Programming and data analysis	MATLAB; Python; R.
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High-performance computing and operating systems Linux; HPC environments.

Environmental and remote sensing Google Earth Engine; QGIS; HEC-RAS; KYPipe; EPANET; SWAT; VSAFT2D.

Architecture and structural analysis AutoCAD; Civil3D.

Writing and productivity LaTeX; Microsoft Office.

LANGUAGES

English IELTS Overall Band Score 7.0; CEFR Level C1.

Portuguese Native.

Spanish Intermediate knowledge.

PROFESSIONAL SERVICE

Ad hoc reviewer Reviewed manuscripts for *Water Resources Research*, *Environmental Modelling & Software*, *Frontiers in Water*, and *Natural Hazards*.

REFERENCES

Peter A. Troch Professor and Department Head, Department of Hydrology and Atmospheric Sciences, University of Arizona, Tucson, Arizona. patroch@arizona.edu. [Website](#); [Google Scholar](#).

Pieter Hazenberg Senior Research Scientist, Applied Research Center, Florida International University, Miami, Florida. phazenbe@fiu.edu. [Website](#); [Google Scholar](#).

Hoshin Gupta Regents Professor, Department of Hydrology and Atmospheric Sciences, University of Arizona, Tucson, Arizona. hoshin.gupta@hwr.arizona.edu. [Website](#); [Google Scholar](#).

Suzana Maria Gico Montenegro Professor, Civil Engineering Department, Federal University of Pernambuco, Recife, Pernambuco. suzana.montenegro@ufpe.br. [Website](#); [Google Scholar](#).

Eduardo Mario Mendonço Professor, Hydraulics and Sanitation Department, University of Sao Paulo, Sao Carlos School of Engineering, Sao Carlos, Sao Paulo. emm@usp.br. [Website](#); [Google Scholar](#).

Maria Mercedes Gamboa Professor, Hydraulics and Sanitation Department, University of Sao Paulo, Sao Carlos School of Engineering, Sao Carlos, Sao Paulo. mmgamboam@usp.br. [Website](#).